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Magnetostratigraphy in the Sydney Basin, Southeastern Australia

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Late Permian and Early Triassic sediments of the Narrabeen Group have been sampled in the coastal section of the Sydney Basin between Wybung Head (the top of the Newcastle Coal Measures to the north of Sydney) and Coal Cliff (above the Bulli Seam south of Sydney). An ancient magnetisation (presumed original) has been isolated only from the red claystone component of the entire section. The magnetisation of the green and buff coloured medium/coarse grained sandstones belonging to the Munmorah Conglomerate and the Tuggerah Formation has been reset during the Late Mesozoic period of magnetic overprinting. This event is primarily responsible for frustrating an attempt to describe properly the magnetostratigraphic record for the Palaeozoic/Mesozoic boundary from deposits in the eastern part of the Sydney Basin. The primary magnetisation of the red beds has survived magnetic reprinting but evidence of overprinting is present during thermal cleaning to 600°C. The detailed magnetostratigraphy of the red beds was revealed after thermal treatment at 650°C and 660°C. The high temperature directions yield a palaeomagnetic south pole for the Patonga Claystone at latitude 30.4°S, longitude 146.9°E, $\alpha_{95} = 7.8°$ and the difference vectors calculated from the low temperature directions yield an overprint south pole position at latitude 57.3°S, longitude 143.4°E, $\alpha_{95} = 3.6°$.

In the light of recent palaeomagnetic investigations in southeastern Australia and the recognition of ubiquitous Late Mesozoic magnetic overprinting, a reassessment of earlier results based upon samples collected at least in the eastern regions of the Sydney Basin may be required.

1. Introduction

The Sydney Basin in southeastern Australia is a major structural unit within which more than 4,800 m of marine and fluvial sediments were deposited, mainly between the Late Carboniferous and the Middle Triassic. It contains the major Permian coal deposits in New South Wales (BRANAGAN *et al.*, 1976). Regional stratigraphic correlation, especially in the fluvial parts of the sequence, is severely hampered by the time transgressive nature of basinal lithofacies and the scarcity of continuous and reliable time parallel horizons. These factors have hindered our understanding of the interrelationships of the sedimentary units and attempts to reconstruct the depositional history of the basin.

Since the geomagnetic field of the Late Palaeozoic is characterised by a long period of reversed polarity (IRVING and PARRY, 1963; KHRAMOV, 1967), identification of the transition which marks the beginning of the Early Triassic mixed interval (e.g. HELSLEY, 1969) will potentially provide a time horizon suitable for intra- and inter-basin correlation. The term Illawarra Reversal was proposed by IRVING and PARRY (1963) for the transition which they originally predicted upon the basis of early Triassic normal magnetisation (IRVING, 1963) following the late Palaeozoic reversed magnetisation measured in igneous and sedimentary rock units within the Sydney Basin.

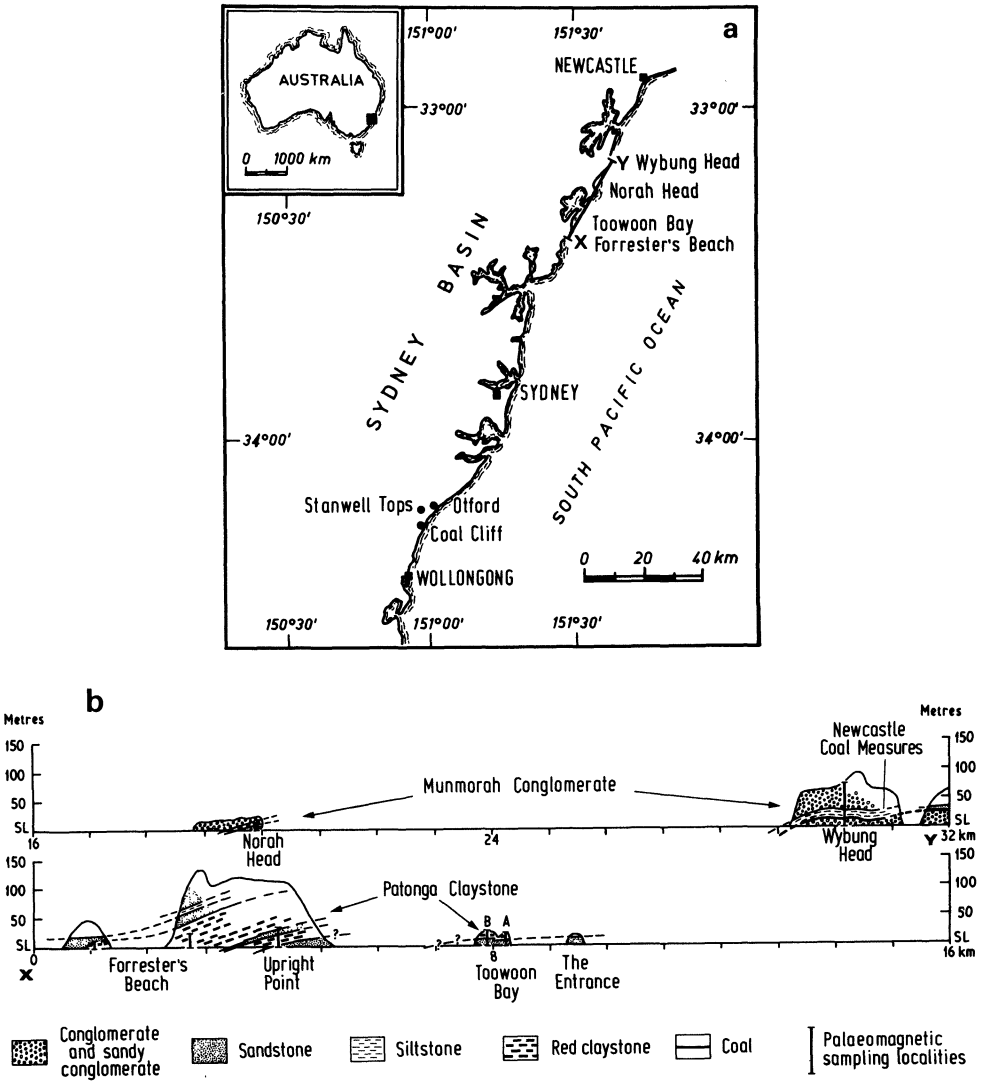


Fig. 1. a) Locality map. Coastal section XY is shown in Fig. 1b at an enlarged scale. b) Coastal section XY showing palaeomagnetic sampling localities. Total length of section 32 km $V:H=10:1$.

A project has therefore been commenced to investigate the magnetostratigraphy of sections within the basin with a view to establishing correlation guidelines based on the geomagnetic field reversal stratigraphy. Detailed sampling within the Late Permian-Middle Triassic Narrabeen Group has been carried out north and south of Sydney at coastal outcrops between Wybung Head and Coal Cliff (Fig. 1). The beds are essentially flat lying, regional dips of 1 to 2° probably being the result of Tertiary warping.

The lower units of the Narrabeen Group in the coastal section north of Sydney are characterised by conglomerates and coarse grey, green, and buff coloured sandstones. The sediments were carried by streams and deposited in channels located between flood plain fine sands and thin bands of red claystone formed in part by overbank deposition. Higher in the section a relatively persistent red claystone horizon is encountered; in the northern coastal section this is known as the Patonga Claystone and a similar horizon (though possibly younger) south of Sydney is the Bald Hill Claystone (HERBERT, 1970).

2. Reversal Stratigraphy

Stable magnetic remanence measured in all sedimentary types other than the red claystones did not survive thermal treatment above 300°C. The demagnetisation paths in the vector diagrams (Fig. 2) demonstrate this behaviour. Clearly, the elucidation of characteristic primary components of magnetic remanence in these sediments, if retained, will depend upon combined application of suitable cleaning methods; this is to be investigated.

In contrast, the magnetic remanence measured in red claystones was stable to thermal demagnetisation to 660°C. The detailed reversal stratigraphy was revealed only after treatment above 600°C. The vector demagnetisation diagrams (Fig. 2) show the decay of two principal components. The high temperature directions yield the reversal patterns shown in Fig. 3. The pattern obtained is independent of diagenetic features and the transitions do not coincide with lithological boundaries. The metre scales for each section show the thickness of section sampled in each case.

The lowest red claystone bands encountered are at Norah Head. They occur between 200 and 300 m below the base of the Patonga Claystone and are interbedded with the higher beds of the basal unit, the Munmorah Conglomerate, of the Narrabeen Group. The Permian-Triassic boundary probably lies within the Munmorah Conglomerate (HELBY, 1973). The transition from reversed to normal polarity represents the oldest reversal encountered.

The base of the Patonga Claystone crops out close to sea-level at Toowoon Bay. The 16 m section here reveals a dominantly reversed to normal polarity pattern and there are two disturbed zones (normal events?) within the lower reversed section. Moving south towards Sydney and higher in the succession, the section labelled Forrester's Beach in Fig. 3 yields a similar reversal stratigraphy. If the base of this section, sampled at Upright Point, lies stratigraphically above the top of the Toowoon Bay section a transition from normal to reversed polarity has been missed. There is a total absence of outcrop between those two localities—approximately 3 km apart. Normal magnetisation persists to the top of the Patonga Claystone, sampled at the south end of Forrester's Beach.

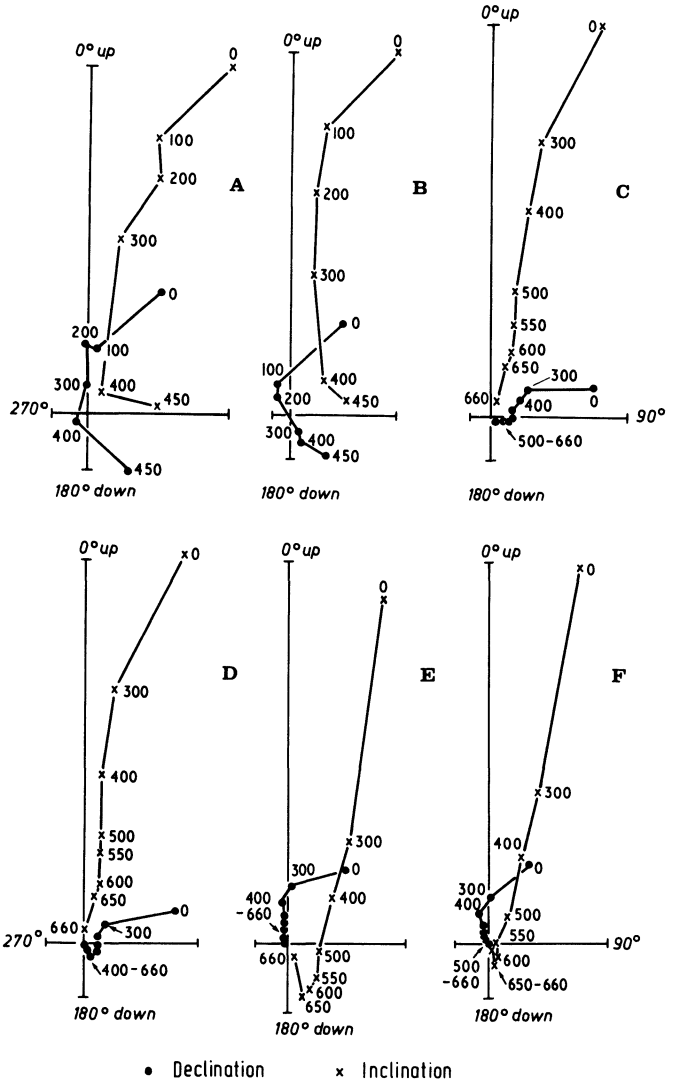


Fig. 2. Vector demagnetisation diagrams. Declination (dots) and inclination (crosses) projected onto the horizontal and vertical planes shown directly at each demagnetisation step. A and B are examples of the intermediate to high temperature magnetic remanence instability in sandstones; C and D examples of primary-normal red claystone; E and F examples of primary-reversed red claystone.

To the south of Sydney, the Bald Hill Claystone was sampled at two road-cut localities about 1 km apart, each covering 3 m of section. The results are summarised in Table 1. In both cases the polarity pattern shows a reversed interval followed by a normal interval.

Although it is not unreasonable to suggest a correlation between the two sections south of Sydney, it remains premature to extend the correlation to the north coast

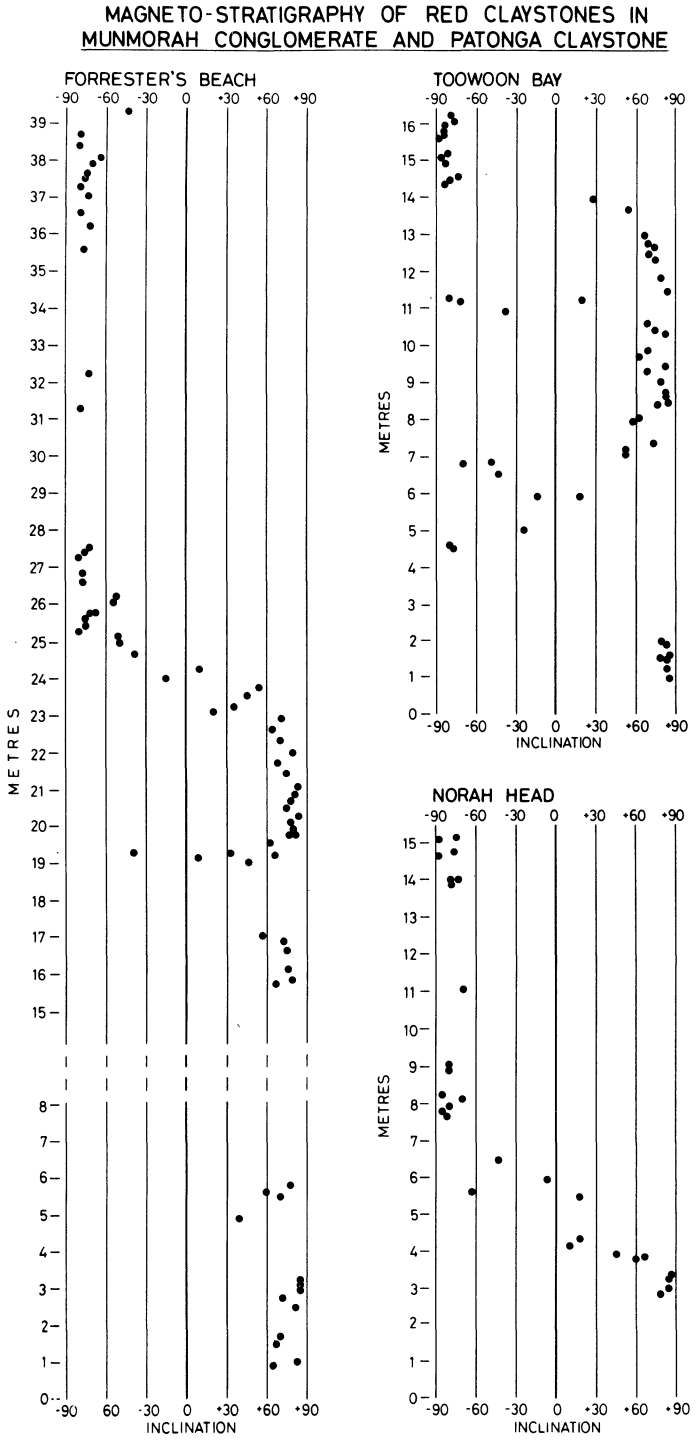


Fig. 3. Magnetic polarity sequences in the red claystones.

Table 1. Bald Hill Claystone reversal stratigraphy. Inclinations measured after treatment of 600°C.

Road-cut locality		Otford		Stanwell Tops	
Arbitrary level		Inclination of magnetisation	Arbitrary level		Inclination of magnetisation
3.0 m		−72°	3.0 m		−76°
2.85 m		−86°	1.7 m		−78°
2.5 m		−76°	0.6 m		+69°
2.4 m		−10°	0.45 m		+41°
0.5 m		−15°	0.0 m		+42°
0.3 m		+28°			
0.2 m		+64°			
0.1 m		+69°			
0.0 m		+89°			

sections. The preferred geological viewpoint would be that the Bald Hill Claystone lies stratigraphically above (and is probably younger than) the Patonga Claystone. It is well-known that the Early Triassic is characterised by frequent reversals of the geomagnetic field (e.g. HELSLEY and STEINER, 1974) and in these circumstances the repeated occurrence of a reversed to normal transition is insufficient evidence in itself for establishing correlation.

The behaviour of the geomagnetic field during the transitions has also been investigated. The intensity data have been grouped according to the polarities obtained after high temperature treatment. Because the normal and reversed directions are near-vertical, the transitional averages were estimated from those specimens whose inclinations of magnetisation lay between +60° and −60°. The average intensities (and standard errors of the means) calculated after the removal of the overprint components were: for the normal intervals of the Forrester's Beach and Toowoon Bay sections, $44.05 \pm 5.0\,\mu$G and $33.0 \pm 7.0\,\mu$G respectively; for the reversed intervals, $42.0 \pm 5.0\,\mu$G and $25 \pm 1.0\,\mu$G respectively; and for the transitional intervals, $3.0 \pm 0.7\,\mu$G and $4.4 \pm 1.3\,\mu$G respectively. The intensities are lower in the transitions by one order of magnitude. The data are interpreted to indicate that the polarity intervals result from reversals of the geomagnetic field and that, during the transitions, the magnetising field was correspondingly weak. There is no clear signature within the directional information that is repeated and upon which correlation can be suggested (cf. KHRAMOV and PETROVA, 1972, p. 332).

3. Palaeomagnetic Directions and Poles

Early work on the red claystones (the Narrabeen Chocolate Shales) by IRVING (1963) involved thermal demagnetisation to only 300°C. At that temperature all directions remained normal, and it was on this basis that the Illawarra Reversal was proposed. All 300°C directions from this study (Table 2), except for a 2 m section at Toowoon Bay, also remained normally magnetised. The measured components at this temperature in fact represent a complex multi-component magnetisation comprising a stubborn normal overprint and a weak normal and reversed primary component. At 300°C a present field

Table 2. 300°C directions in the Patonga Claystone and lower red claystones of the Narrabeen Group.

Section	N	Dec°	Inc°	R	α_{95}°
Norah Head	19	014.0	−79.0	18.706	4.3
Toowoon Bay A	7	327.5	+79.0	6.322	21.5
B	39	355.5	−66.0	36.253	13.8
Upright Point	13	006.0	−62.0	12.876	4.2
Forrester's Beach					
North	49	024.5	−74.0	47.916	3.0
South	15	054.5	−79.0	14.352	8.4
Otford	17	352.5	−77.5	16.359	7.2
Mean	6	010.5	−73.5	5.935	7.7
IRVING (1963)	4 (sites)	336.5	−82.0	3.965	10.0

N=number of samples, Dec°=declination of magnetic remanence, Inc°=inclination of magnetic remanence, R=resultant of N unit vectors, α_{95}°=semi-angle of the cone of confidence calculated at the 95% probability level (FISHER, 1953).

viscous remanent magnetisation has possibly been eliminated.

Difference vectors calculated after incremental thermal demagnetisation treatment between 200°C and 550°C revealed the persistent removal of the overprint component in all red claystone specimens from the north coast. These results are summarised in Table 3. The high temperature components of magnetisation with mixed polarities (regarded here as primary) have been analysed with respect to polarity within each section (Table 4). They yield a palaeomagnetic pole for the Early Triassic at latitude 30.4°S, longitude 146.9°E, and $\alpha_{95}=7.8°$. The overprint directions yield a pole position at latitude 57.3°S, longitude 143.4°E, and $\alpha_{95}=3.6°$.

The overprint pole lies on the Late Cretaceous section of the apparent polar wander curve for Australia (EMBLETON, 1980). The severity of magnetic overprinting has hindered earlier attempts to define accurately the configuration of the Mesozoic geomagnetic field from palaeomagnetic studies in southeastern Australia: this problem has been discussed more fully by SCHMIDT (1976a, b).

4. Concluding Remarks

Magnetostratigraphy will no doubt play an important role in establishing correlation guidelines within the Sydney Basin. Clearly this requires the development of palaeomagnetic techniques suitable to the variety of rock types present. In addition, the identification of the Late Cretaceous magnetic overprint has important implications for models concerned with the geological evolution of the basin, particularly in relation to the timing of formation of its rifted eastern margin. The overprint was probably acquired during a period of burial and subsequent uplift caused by the removal of overburden at the time of initial rifting.

The Early Triassic palaeomagnetic pole position is different from the pole reported by IRVING (1963). The reasons for this have been outlined above and consequently we may

Table 3. Difference vectors averaged by section in the red claystones of the Munmorah Conglomerate and the Patonga Claystone. Removed vectors shown to be uniformly directed irrespective of polarity.

Section	Normal magnetisation					Transition					Reversed magnetisation				
	N	Dec°	Inc°	R	α_{95}°	N	Dec°	Inc°	R	α_{95}°	N	Dec°	Inc°	R	α_{95}°
Norah Head	12	013.8	−82.1	11.882	4.5	—	—	—	—	—	4	356.4	−77.0	3.991	4.9
Toowoon Bay	13	006.2	−77.3	12.949	2.6	9	349.6	−77.4	8.959	3.7	17	023.2	−76.2	16.912	2.6
Forrester's Beach North	14	025.0	−76.8	13.808	4.8	7	010.9	−76.8	6.966	4.5	15	013.6	−75.2	14.936	2.5
Otford	7	013.8	−76.5	6.867	9.0	—	—	—	—	—	9	009.0	−73.9	8.829	7.5
Mean (section unit weight)	4	014.8	−78.2	3.996	3.5	—	—	—	—	—	4	010.7	−75.7	3.996	3.3

Mean direction (unit weight to polarity intervals and transition sections):

N	Dec°	Inc°	R	α_{95}°
10	010.2	−77.1	9.986	1.9

See footnotes to Table 2.

Table 4. High temperature directions (660°C) averaged by section for polarity intervals in the red claystone of the Munmorah Conglomerate and the Patonga Claystone.

Section	Normal magnetisation					Reversed magnetisation				
	N	Dec°	Inc°	R	α_{95}°	N	Dec°	Inc°	R	α_{95}°
Norah Head	14	083.4	−88.0	13.825	4.6	—	—	—	—	—
Toowoon Bay	13	017.9	−82.9	12.971	2.0	19	317.0	+82.8	18.707	4.3
Upright Point	—	—	—	—	—	11	341.1	+82.9	10.736	7.5
Forrester's Beach										
North	10	125.0	−83.8	9.921	4.5	14	051.6	+86.1	13.748	5.5
South	13	019.3	−88.8	12.914	3.5	—	—	—	—	—
Mean (section unit weight)	4	064.6	−87.3	3.990	5.3	3	343.9	+84.9	2.994	7.0

Mean direction irrespective of polarity:

N	Dec°	Inc°	R	α_{95}°
7	125.5	−87.5	6.974	3.9

See footnotes to Table 2.

need to critically review certain other results obtained previously for the Late Palaeozoic from southeastern Australia. Although field and laboratory stability tests carried out by IRVING (1966) on Carboniferous rocks from New South Wales would seem to have established the position of the Late Carboniferous pole beyond doubt, further work is required to investigate the configuration of the geomagnetic field during Permian time.

The field sampling programme was assisted by Dr. J. Kirschvink, J. Nolan, and B. Taylor. Dr. Kirschvink was also involved with the analysis of results. All measurements were carried out in the palaeomagnetic laboratory of the Research School of Earth Sciences, Australian National University, with a cryogenic magnetometer kindly loaned by Dr. E. M. Shoemaker of the California Institute of Technology.

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